
Enterprise Production Guide v1.0

Repository: Omegalflop-Intern / ERP_Management

Architecture: Node.js Express + React 18 Vite + MongoDB

Modules: 26 Enterprise Modules

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1. Executive System Overview

Brothers Mobile Shop ERP is an end-to-end Enterprise Resource Planning (ERP) platform built specifically for mobile phone retail and wholesale businesses. It seamlessly connects sales, inventory stock, IMEI serial tracking, HR payroll, multi-branch operations, customer relationship management (CRM), double-entry accounting, and repair job management into a single real-time dashboard.

System Infrastructure & Access URLs

- **Client Web Application:** <http://localhost:3000> (React 18 + Vite 5 + Tailwind CSS)
- **Backend REST API Server:** <http://localhost:5000/api/v1> (Node.js 20 + Express 4 ESM)

- **Interactive Swagger API Docs:** `http://localhost:5000/api/docs`

- **Database Connection:** MongoDB 7 (`mongodb://127.0.0.1:27017/mobile_shop_erp`)

Default Access Credentials

Role	Default Identifier / Username	Default Password setup
System Administrator	<code>admin</code>	Set via <code>SEED_PASSWORD_ADMIN</code> in <code>server/.env</code>
Branch Manager	<code>manager</code>	Set via <code>SEED_PASSWORD_MANAGER</code> in <code>server/.env</code>

2. How to Run & Access the System

A. Quick Start (Development Mode)

To run the system locally, start the Express backend server and React frontend client in separate terminals:

Terminal 1: Start Backend Server (Port 5000)

```
cd server && npm install && npm run seed && npm run dev
```

Terminal 2: Start Frontend Client (Port 3000)

```
cd client && npm install && npm run dev
```

B. Docker Container Deployment

For containerized deployment with automated MongoDB, Backend, and Client containers:

```
docker compose up -d --build
```

3. Core Workflows & Data Flow

Workflow A: Sales & IMEI Stock Deduction

1. Cashier opens POS screen on `http://localhost:3000/dashboard/sales`.
2. Scans or searches product by IMEI serial number or barcode.
3. System verifies IMEI status (must be `In Stock`).
4. Order is processed, generating invoice PDF, updating sales revenue, updating customer due/balance, and automatically marking IMEI status as `Sold`.

Workflow B: Device Repair Management

1. Technician logs incoming faulty device on Repair module.
2. System generates Repair Job Sheet with initial issue diagnosis and estimated cost.
3. Device status updates (`Received` → `In Progress` → `Repaired` → `Delivered`).
4. Upon delivery, repair invoice is generated and linked to double-entry accounting income.

Workflow C: HR Attendance & Payroll

1. Employees check-in / check-out daily.
2. System computes total working hours, overtime, and leave deductions.
3. At month-end, Payroll service calculates gross salary, allowances, and generates printable salary slips.

4. Complete API Endpoints Catalog (26 Modules)

Below is the complete reference of all REST API modules exposed at `http://localhost:5000/api/v1`:

Module	Endpoint Base Route	Key Operations & Capability
Auth	<code>/auth</code>	Login (<code>/login-direct</code>), OTP verification, Password Reset, Logout
User & Roles	<code>/users</code> , <code>/roles</code>	User CRUD, Role-Based Access Control (RBAC) permissions matrix
Catalog	<code>/catalog</code>	Categories, Brands, Models, Unit of Measurements
Product	<code>/products</code>	Product catalog, prices, variants, cost margin analysis
IMEI Tracker	<code>/imei</code>	Serial number tracking, status updates (In Stock, Sold, Repaired), history
Stock & Inventory	<code>/stock</code>	Stock quantities, low-stock alerts, branch stock transfers
Supplier	<code>/suppliers</code>	Supplier directory, outstanding dues, payment history
Purchase	<code>/purchases</code>	Purchase orders, incoming stock intake, purchase invoices
Sale (POS)	<code>/sales</code>	Retail sales, invoice generation, thermal print receipts, sales returns
Wholesale	<code>/wholesale</code>	Bulk wholesale orders, customized tier pricing, bulk shipping
Customer (CRM)	<code>/customers</code>	Customer profiles, due collections, purchase history
Warranty	<code>/warranties</code>	Warranty claims processing, policy verification, replacements

Module	Endpoint Base Route	Key Operations & Capability
Repair Service	/repairs	Repair job sheets, technician assignments, repair status tracking
Accounting	/accounting	Chart of accounts, journal entries, Trial Balance, P&L, Balance Sheet
Expense	/expenses	Operational expense logging, expense categories, vouchers
Investor	/investors	Investor capital tracking, profit-sharing distribution calculation
Loan	/loans	Business loan records, interest computation, repayment schedules
Employee	/employees	Staff profiles, designations, salary structures, documents
Attendance	/attendance	Daily check-in/out, hours computation, attendance logs
Leave	/leaves	Leave applications, multi-level approval workflows, leave balances
Payroll	/payroll	Monthly salary sheet generation, bonus/penalty adjustments, pay slips
Branch	/branches	Multi-branch location management, branch performance tracking
Settings	/settings	Company profile, thermal print templates, design themes, SMTP setup
Notification	/notifications	Real-time alerts, stock warnings, system audit events
Real-time SSE	/sse	Server-Sent Events stream for live notifications & POS updates

Module	Endpoint Base Route	Key Operations & Capability
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Report Analytics /reports Revenue charts, sales analytics, top products, HR summaries

5. UI Design Modes & Interactive Features

The ERP client provides 7 customizable design modes stored in browser `localStorage`:

- **Flat:** Clean, minimal corporate styling.
- **Neumorphism:** Soft, extruded tactile interface.
- **Glassmorphism & Glassmorphism Pro:** Frosted glass backdrop-blur effects with gradient borders.
- **Liquid Glass:** Dynamic translucent fluid design.
- **Neo Brutalism:** High-contrast bold borders with retro offset box-shadows.
- **Aurora:** Vibrant glowing background gradient animations.

6. Backup & Disaster Recovery

System backups can be triggered directly or scheduled:

```
cd server && node scripts/run-restore.js
```